



Human Touch in Digital Learning Communities: Teacher's Role in Shaping Learner Satisfaction on EdTech Platforms

Jinesh Jain^{1*}, Manpreet Kaur¹, Kiran Sood²

¹ Sri Aurobindo College of Commerce and Management, 142021 Ludhiana, India

² Chitkara Business School, Chitkara University, 140401 Rajpura, India

* Correspondence: Jinesh Jain (jineshjain@sacem.in)

Received: 02-10-2026

Revised: 03-09-2026

Accepted: 03-20-2026

Citation: Jain, J., Kaur, M., & Sood, K. (2026). Human touch in digital learning communities: Teacher's role in shaping learner satisfaction on EdTech Platforms. *Cent. Community Dev. J.*, 6(1), 26–39. <https://doi.org/10.56578/ccdj060103>.



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Abstract: This study examines learner satisfaction within digital learning environments by situating EdTech Platforms as emerging community-based learning systems. Drawing on the DeLone and McLean Information Systems Success Model (IS Success Model), the research investigates the effects of information quality, service quality, and system quality on learner satisfaction, while incorporating teacher's role as a moderating factor. Primary data were collected from 473 school students engaged in EdTech Platforms. The findings confirm that all three quality dimensions significantly influence learner satisfaction. Moreover, teacher's role is significant in shaping these relationships: It strengthens the effects of information quality and service quality, while reducing the relative impact of system quality. These results suggest that digital learning outcomes are not determined solely by technological features, but are co-produced through interactions between platform characteristics and human support. By interpreting EdTech Platforms as community-oriented learning environments, the study highlights how teacher involvement contributes to the development of supportive, interactive, and adaptive learning communities. The findings offer implications for the design and governance of digital education systems, particularly in contexts where equitable access, engagement, and collective learning experiences are central to community development.

Keywords: Digital learning communities; EdTech Platforms; Learner satisfaction; Teacher's role; Community-based learning; Information Systems Success Model

1. Introduction

Educational technology (EdTech) has emerged as a robust form of learning thanks to technology usage in education (Williamson, 2021). Substantial increase in online courses and platforms has been witnessed due to the indisputable importance of EdTech in education post COVID-19 period (Al-Fraihat et al., 2020; de Paula, 2021; Dwivedi & Virmani, 2023; Fariz & Winarsih, 2025). As a highly diverse field of research and development, EdTech encompasses institutions, content, technological forms, and discourses involving a wide range of actors (Williamson, 2021). EdTech advocates have argued that digital education can enhance access, learning, and participation (Laufer et al., 2021). Studies have argued that technology can be deployed to improve learning outcomes, increase cost efficiency, and support educational retention and retrieval, among other potential benefits (Hackman & Reindl, 2022).

EdTech interventions centered on self-directed learning and instructional enhancement are among the most effective approaches for improving learning outcomes (Rodriguez-Segura, 2022). Technology can enhance teachers' pedagogy by facilitating the teaching-learning process, improving instructional responsiveness, and strengthening learners' internal motivation through more positive attitudes toward the subject. Studies in the literature have reported that learner satisfaction is significantly influenced by service quality of education system (Darawong & Sandmaung, 2019; Hasan et al., 2008; Huang, 2010; Spreng & Mackoy, 1996). Moreover, teacher-driven support appears to be a crucial factor in improving learning quality (Kiriakidis, 2008; Ladyshevsky, 2013). Furthermore, motivation, interface quality, and teacher role contribute to positive online learning evaluations,

which in turn enhance learner satisfaction (Thanasi-Boçe, 2021).

The DeLone and McLean Information Systems Success Model (IS Success Model) is one of the most widely cited information systems models and has broader applicability in the educational field than many other models of technology acceptance and use. The model has been widely applied in studies of digital learning and information systems, where system quality, information quality, and service quality are commonly treated as core antecedents of user satisfaction and system use. However, empirical findings remain mixed across different contexts. Only a limited number of studies have examined the model specifically in relation to learner satisfaction on EdTech Platforms (Alyoussef, 2023; Rulinawaty et al., 2024). As this model goes beyond user behavior and provides theoretical support for the relationships among system quality, service quality, and information quality in the e-learning environment, it is reasonable to extend it by incorporating teacher role as a moderator in assessing learner satisfaction on EdTech Platforms.

This study contributes to the existing literature on user satisfaction in e-learning environments by extending the 2003 DeLone and McLean IS Success Model and examining the moderating role of teacher role in shaping satisfaction with system quality, service quality, and information quality. In the second section, extensive literature was gone through to formulate the hypotheses examining the constructs of DeLone and McLean model and teacher's role on EdTech Platforms. The third section presents research methodology followed by data analysis, findings and discussion. The paper concludes with the implications and limitations of the study.

2. Literature Review

An extensive review of literature was conducted to identify the gap and formulate research hypotheses. This section is divided into three parts, first part deals with studies exploring DeLone and McLean IS Success Model and learner satisfaction with EdTech Platforms, second part studies impact of teacher's role on learner satisfaction and the last part covers studies moderating role of teacher on relationship between DeLone and McLean IS Success Model and learner satisfaction with EdTech Platforms.

2.1 DeLone and McLean IS Success Model and Learner Satisfaction with EdTech Platforms

The DeLone and McLean IS Success Model was originally proposed by DeLone and McLean in 1992. It synthesized previous research on information systems into a more coherent body of knowledge. Its primary purpose was to identify the key factors that determine information systems success. The model and the relationships among its constructs have been validated by numerous empirical studies. However, several studies suggested improvements to the original model. In response to these recommendations, DeLone and McLean revised and extended the model in 2003. The updated model conceptualizes information system quality in terms of three dimensions: service quality, system quality, and information quality. The model has also been applied in broader digital contexts, including e-commerce (DeLone & McLean, 2004).

2.1.1 Information quality

Information quality consists of two elements, namely content adequacy (i.e., reliability, adequacy, and completeness of the information provided) and content usefulness (i.e., informativeness and suitability of the information presented) (Nguyen & Tran, 2018). Amidst the ever-increasing extent of information that is accessible on the internet, the quality of information turns out to be a vital challenge, for internet users, administrators, content providers, webmasters, and IT-staff (Eppler et al., 2003) apart from the EdTech teachers who need to assure the quality of published content. The quality of information signals available to a user during an initial instructor session can help build favorable beliefs and reduce perceived exchange risk (Nicolaou & McKnight, 2006). Information quality has also been found to influence continuance intention through performance expectancy (Lee et al., 2019). Alyoussef (2023) further found that information quality affects both perceived ease of use and perceived usefulness of the systems. Studies have also revealed that perceived information quality contributes significantly to user satisfaction (Eppler et al., 2003; Jun & Kang, 2013; Kim & Lim, 2001). Accordingly, it is hypothesized that Information Quality has significant positive impact on learner satisfaction with EdTech Platforms (H1).

2.1.2 Service quality

Service quality in e-learning includes the quality of course materials, the teacher's role on the e-learning platform, and the quality of administrative support (Pham et al., 2019). Studies pertaining to service quality have been confined to traditional physical teaching environment only. While majority of studies have found significant positive impact of service quality on student satisfaction (Darawong & Sandmaung, 2019; Hasan et al., 2008; Huang, 2010; Spreng & Mackoy, 1996), studies focusing on service quality have also considered the role of values in the association between service quality and learner satisfaction (Caruana et al., 2000; Oh, 1999). Few studies have also focused on the impact of service quality of education on students' loyalty and satisfaction (Chandra et

al., 2018; DOAN, 2021; Mulyono et al., 2020; Pham et al., 2019). Annamdevula & Bellamkonda (2016) in their study confirmed significant impact of service quality on student loyalty with the mediating role of learner satisfaction. In addition to this, studies pertaining to impact of service quality on consumer satisfaction also revealed significant positive relationship between the same (Alexandris et al., 2004; Melastri & Giantari, 2019). Hence, in the present study, we hypothesized that service quality has significant positive impact on learner satisfaction with EdTech Platforms (H2).

2.1.3 System quality

System quality refers to the extent to which a system possesses desirable characteristics that enhance its overall performance (Dreheeb et al., 2016). System quality is a significant success characteristic that affects user satisfaction and intention to use (Delone & McLean, 2003). Studies have associated system quality with satisfaction (Alyoussef, 2023; Bhasin & Rehman, 2025; Rulinawaty et al., 2024). While majority of the studies reported that system quality significantly influences user satisfaction (Jing & Yoo, 2013; Salameh et al., 2018; Tajuddin et al., 2013), the findings of Song et al. (2017) contradicted these results. They examined building information modeling (BIM) user satisfaction in architecture, engineering, and construction (AEC) industries over a survey of BIM operators from China and found that system quality didn't have a considerable influence on the satisfaction. Chang et al. (2012) found positive impact of system quality on service quality, job satisfaction, and system performance in a study conducted in BMC Medical informatics. Also, Gürkut & Nat (2017) revealed that system Quality has indirect significant impact on system satisfaction in Higher Education Institutions. Due to non-existence of conclusive evidence, in the present study, it has been hypothesized that System Quality has significant positive impact on learner satisfaction with EdTech Platforms (H3).

2.2 Teacher's Role and Learner Satisfaction with EdTech Platforms

The existing was reviewed to examine studies that have analyzed the relationship between teacher's role and learner satisfaction. Lee et al. (2018) confirmed the relationship between learning-environment qualities, teacher involvement and learner satisfaction, while Micari & Pazos (2016) revealed that teacher connectedness affects student satisfaction, and that self-efficacy functioned as a mediator between both learner orientation and teacher connectedness on the one hand, and satisfaction on the other. Dwivedi & Virmani (2023) also signified the role of teachers in an online set up and concluded that teacher, by motivating and engaging learner, can create a learning environment and hence increase learner satisfaction. Dennen et al. (2007) reported that learner performance was more closely tied to teacher actions focused on course content, whereas learner satisfaction was more strongly associated with students' perceptions that their interactive communication needs were met. Aldholay et al. (2018) concluded that leadership role played by a teacher significantly impacts usage of online learning system by students. This construct needs further exploration due to dearth of available literature. Hence, it is hypothesized that Teacher role has significant positive impact on learner satisfaction with EdTech Platforms (H4).

2.3 Moderating Role of Teacher on Relationship between DeLone and McLean IS Success Model and Satisfaction with EdTech Platforms.

Teachers play a crucial role in the success or failure of EdTech systems. Their proficiency and application of technology instruments in online teaching affect the quality of information disseminated, their system use and satisfaction during course delivery (Gay, 2016). Aldholay et al. (2018) concluded that leadership role played by a teacher significantly mediates the relationship between overall quality of online learning system and students' satisfaction. Although there is dearth of literature examining the impact of teacher role on information quality, but Lee et al. (2018) in their study revealed that the teacher involvement had a moderating effect on flexible contents qualities, the results of which was in consonance with the findings of study conducted by Lala et al. (2002), which revealed a strong effect related to predilection for a superior information quality seal.

Researchers have long examined aspects of system quality that contribute to student satisfaction in both traditional and modern online learning environments. Teacher role in users' perceptions of system quality has been identified as an important factor shaping learner satisfaction. Several studies have also identified it as a significant predictor of satisfaction. Studies have demonstrated it as a significant predictor of satisfaction. (Al Mulhem, 2020; Koh & Kan, 2020; Lee et al., 2018). Almarashdeh (2016) emphasized the importance of designing learning management systems based on the needs of both teachers and students while incorporating the latest technologies. Few studies have examined learner satisfaction in massive open online courses (MOOCs) settings from the perspective of platform and course quality, although direct evidence on the teacher's moderating role remains limited (Hew et al., 2020). Hence, due to lack of conclusive evidence of role of teacher in affecting user satisfaction with service, system, and information quality of EdTech Platforms, following hypotheses have been framed:

H5: Teacher's role moderates the relationship between information quality and learner satisfaction with EdTech Platforms

H6: Teacher's role moderates the relationship between service quality and learner satisfaction with EdTech Platforms

H7: Teacher's role moderates the relationship between system quality and learner satisfaction with EdTech Platforms

The direction and nature of the moderating effect of teacher role on the relationships among information quality, service quality, system quality, and learner satisfaction remain unclear. Existing studies have not yet provided conclusive evidence regarding whether this moderating effect is positive or negative. Moreover, most prior research has been conducted in developed countries, leaving emerging markets relatively underexplored. Accordingly, this study extends the DeLone and McLean IS Success Model by incorporating teacher role and examining its effect on learner satisfaction in the context of EdTech Platforms.

3. Methodology

3.1 Overview of the Proposed Research Model

Figure 1 depicts the proposed model. The proposed model is based on IS Success Model given by DeLone and McLean (DeLone & McLean, 2003). This model states that information quality, service quality, and system quality affect the user satisfaction. Teacher's role in user's perception regarding online learning has been substantiated as the key to learner satisfaction (Al Mulhem, 2020; Koh & Kan, 2020; Lee et al., 2018). Therefore, the basic model proposed by Delone & McLean (2003) has been extended by incorporating teacher role as a moderating variable. Seven hypotheses have been tested in the present research based on the proposed model.

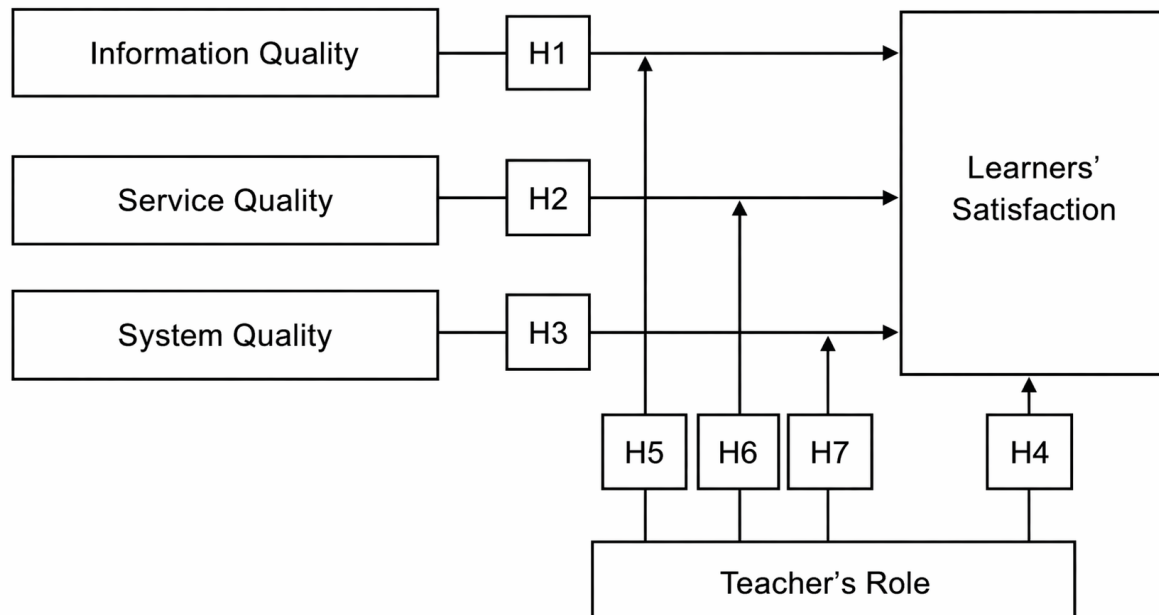


Figure 1. Research framework

3.2 Development of Instrument

An 18-item survey instrument was established for this study where items were measured on a 5-point Likert Scale from Strongly Disagree to Strongly Agree. Table 1 reveals the variables of the study and their source. To increase the authenticity of the questionnaire, pre-testing was done with 50 respondents chosen at random from the population. The results of pre testing showed that Cronbach's alpha values for all items were more than 0.70 showing reliability of the questionnaire.

3.3 Data Collection

Primary data were collected from India by means of a self-structured questionnaire administered through Google Forms during the period January 2024 till June 2025. Questionnaire was distributed to the students studying in 5th to 12th standard through the teachers teaching on EdTech Platforms who shared the same in their WhatsApp groups. Snowball sampling technique was employed for collecting data because it was not possible to contact the students who had used or were using EdTech Platforms. Earlier research has confirmed that when respondents are

from a specific population (Nardi, 2018) and target population is not easily accessible (Wagner et al., 2014), and is interconnected through a network, snowball sampling method is the most appropriate sampling technique. Hence, in the present study, this technique has been used as the nature of population is specific (i.e., students studying between fifth to twelfth standard using EdTech Platforms) and they belong to an interconnected network of teachers teaching at such platforms. Out of the total responses received, 473 responses were found to be useful for the final analysis. Remaining questionnaires were incomplete, and they were not used for the analysis.

Table 1. Variables and their sources

Constructs		Source	Code	Items
System quality		Adapted from Alotaibi (2021)	SY1	My EdTech platform is user friendly
			SY2	My EdTech platform allows interaction between user and platform
			SY3	My EdTech platform has high speed information access
Information quality	Delone and McIan Information Systems Success Model	Adapted from Aldholay et al. (2018)	IF1	Information (like audios, videos etc) available at EdTech platform is accurate
			IF2	Information available at EdTech platform is up to date
			IF3	Information available at EdTech platform is organised
Service quality		Adapted from Aldholay et al. (2018)	SEQ1	EdTech platform provides proper online assistance
			SEQ2	EdTech platform's functionality is up to the mark
			SEQ3	EdTech platform provides good interactivity
			SEQ4	EdTech platform provides rapid response to my service requests
			SEQ5	EdTech platform provides quick solution to my complaints
Teacher's role		Adapted from Aldholay et al. (2018)	IR1	Teacher enthusiasm about EdTech platform stimulates my desire to learn
			IR2	Teacher provides prompt response to my queries
			IR3	Teacher pays individualised consideration to students
			IR4	Teacher inspires me to put additional efforts in the subject
Satisfaction		Adapted from Alotaibi (2021)	SAT1	I am satisfied with performance of my EdTech platform
			SAT2	My EdTech platform helps me in attaining my educational goals
			SAT3	Overall, I am delighted with the experience of utilizing EdTech platform

3.4 Sample Description

Table 2 presents sample description for the present study.

Table 2. Participants characteristics ($n = 473$)

Variables	Category	Frequency	Percentage
Gender	Female	287	60.67
	Male	186	39.33
Class/Standard	Fifth	15	3.17
	Sixth	30	6.34
	Seventh	32	6.76
	Eighth	35	7.39
	Ninth	133	28.11
	Tenth	138	29.17
	Eleventh	24	5.07
	Twelfth	66	13.95
	Educom	125	26.43
EdTech Platforms	Cuemath	114	24.10
	Vedantu	65	13.74
	Byjus	169	38.67

Note: n = sample size

3.5 Analysis Methods

The research study used PASW Statistics (version 18) and partial least squares structural equation modeling (PLS-SEM) for data analysis and hypotheses testing. PASW was used for descriptive analysis and for analysis of common method bias. PLS-SEM was used to analyze the reliability and validity of the scale as well as for

hypotheses testing as it can analyze complex models comprised of many indicators (Hair et al., 2019). This technique is considered suitable in situations where researchers want to develop a new theory or identify key driver constructs (Hair et al., 2016; Henseler et al., 2016).

3.6 Data Analysis and Interpretation

3.6.1 Common method bias

As a single instrument was used to collect data for all independent and dependent constructs, common method bias (CMB) problem may be presented in data which needs to be addressed (Kock, 2015). Podsakoff et al. (2012) have recommended some remedies to deal with problem of CMB which were adapted in the study. First of all, items were properly framed and rechecked to get accurate answers from the respondents. Secondly, Harman single factor test was used to analyze CMB. All the items measuring independent and dependent constructs were loaded onto a single factor through exploratory factor analysis that explained 43.871% of total variance which is less than 50% suggesting that CMB is not an issue in the data (Streukens et al., 2017).

3.6.2 Construct reliability and validity

When the data for multiple variables measuring a construct is collected through a single instrument, multicollinearity can be an issue in statistical analysis. Tabachnick & Fidell (2007) referred to multicollinearity as a situation where there are high correlations among variables in the data set. Hence to check multicollinearity, Variance Inflation factor (VIF) values were calculated for all the items and were found to be in the acceptable range of 0 to 5 (Diamantopoulos & Siguaw, 2006).

Table 3 displays the outer loadings, Cronbach alpha, average variance extracted (AVE) and composite reliability for the outlined constructs. All the values for outer loadings are above the threshold value of 0.70. Reliability was confirmed, as the values of Cronbach's alpha and composite reliability for all constructs were greater than 0.70. Convergent validity was measured using the value of AVE, which was calculated by averaging the square of outer loadings for each construct, the value of which should be more than 0.5 (Fornell & Larcker, 1981). Findings confirmed the convergent validity of the scale.

Table 3. Construct reliability and validity

Items	Information Quality	Service Quality	System Quality	Teacher's Role	Satisfaction
IF1	0.925				
IF2	0.951				
IF3	0.937				
SEQ1		0.816			
SEQ2		0.807			
SEQ3		0.759			
SEQ4		0.837			
SEQ5		0.854			
SY1			0.821		
SY2			0.815		
SY3			0.868		
IR1				0.879	
IR2				0.909	
IR3				0.903	
IR4				0.911	
SAT1					0.873
SAT2					0.852
SAT3					0.837
Cronbach's α	0.931	0.873	0.787	0.922	0.816
Dijkstra-Henseler's rho_A	0.931	0.875	0.819	0.925	0.824
Composite reliability	0.956	0.908	0.873	0.945	0.89
Average variance extracted (AVE)	0.879	0.665	0.697	0.811	0.73

3.6.3 Assessing discriminant validity

First of all, Heterotrait-Monotrait (HTMT) ratios were calculated which are displayed in Table 4. All HTMT values were less than the acceptable value of 0.9 (Ringle et al., 2014). Further Fornell-Larcker criterion was used to check discriminant validity and the results are displayed in Table 5. Bivariate correlations were compared with

square root of AVE for every construct. All the correlations were less than the square root of AVE showing presence of discriminant validity.

Table 4. Heterotrait-Monotrait (HTMT) ratio

Items	Information Quality	Teacher's Role	Satisfaction	Service Quality
Teacher's role	0.704			
Satisfaction	0.602	0.74		
Service quality	0.402	0.53	0.737	
System quality	0.236	0.343	0.446	0.41

Table 5. Fornell-Larcker criterion

Items	Information Quality	Teacher's Role	Satisfaction	Service Quality	System Quality
Information quality	0.938				
Teacher's role	0.655	0.901			
Satisfaction	0.526	0.643	0.854		
Service quality	0.363	0.478	0.631	0.815	
System quality	0.199	0.308	0.372	0.35	0.835

Note: AVE = average variance extracted. Values in bold represent the square root of AVE.

3.6.4 Model and hypotheses testing

A bootstrapping procedure was run in SmartPLS with 5,000 subsamples to examine the hypothesized structural relationships. Table 6 and Figure 2 display the results of the analysis.

Table 6 presents the path coefficients, *t*-statistics, *p*-values, and the results of hypothesis testing. All the constructs of IS Success Model namely, information quality, service quality and system quality were found to be positively influencing satisfaction where service quality ($b = 0.355$, p -value < 0.001) had medium size effect on satisfaction whereas information quality ($b = 0.131$, p -value < 0.001) and system quality ($b = 0.145$, p -value < 0.001) were found to have small size effect on satisfaction. These findings support H1, H2 and H3. As far as H4 is concerned, teacher's role was found to have significant positive impact on satisfaction ($b = 0.344$, p -value < 0.001) statistically supporting H4. The results also supported H5, H6 and H7 as the relationships between all the constructs of IS Success Model and satisfaction was moderated by the teacher role.

Table 6. Path analysis

Hypothesis	Path	Path Coefficient	<i>t</i> -Statistics	<i>p</i> -Value	Decision
H1	Information quality → Satisfaction	0.131	3.602	0.000	Supported
H2	Service quality → Satisfaction	0.355	7.941	0.000	Supported
H3	System quality → Satisfaction	0.145	3.574	0.000	Supported
H4	Teacher's role → Satisfaction	0.344	8.236	0.000	Supported
H5	Teacher's role × Information quality → Satisfaction	0.105	3.167	0.002	Supported
H6	Teacher's role × Service quality → Satisfaction	0.176	5.822	0.000	Supported
H7	Teacher's role × System quality → Satisfaction	-0.223	7.924	0.000	Supported

Figure 3 shows the slope plot which confirms that teacher role entails a strong relationship between information quality, service quality & system quality, and satisfaction with EdTech Platforms as the lines are not parallel. Detailed examination of path coefficients signified that interaction effects of teacher's role*information quality ($b = 0.105$, p -value = 0.002) and teacher's role*service quality ($b = 0.176$, p -value < 0.001) were found to be positive whereas for teacher's role*system quality ($b = -0.223$, p -value < 0.001) was found to be negative and significant. Hence, our results confirmed that teacher's role is significant in influencing satisfaction of students with EdTech Platforms. In addition, teacher's role significantly moderates the impact of IS Success Model on satisfaction. Model fit indicators are also displayed in Table 6. Chi-square for the model (2029.091), standardized root mean square residual (0.063) and normed fit index (0.712) values depict the fitness of the model. Coefficient of determination ($R^2 = 0.572$) value signified good explanatory power of the model.

To analyze predictive power of the model, PLS predict analysis was run as per guidelines given by Shmueli et al. (2019), and the results are displayed in Table 7. Such analysis is conducted to examine whether researched PLS-SEM model is practically useful for predicting real-world outcomes or it is just fitting the observed data. Values of Q^2_{predict} for PLS model were greater than zero for every item of endogenous construct, learner satisfaction,

indicating that the model is capable of predicting learner satisfaction better than a naïve benchmark model that relies only on mean values.

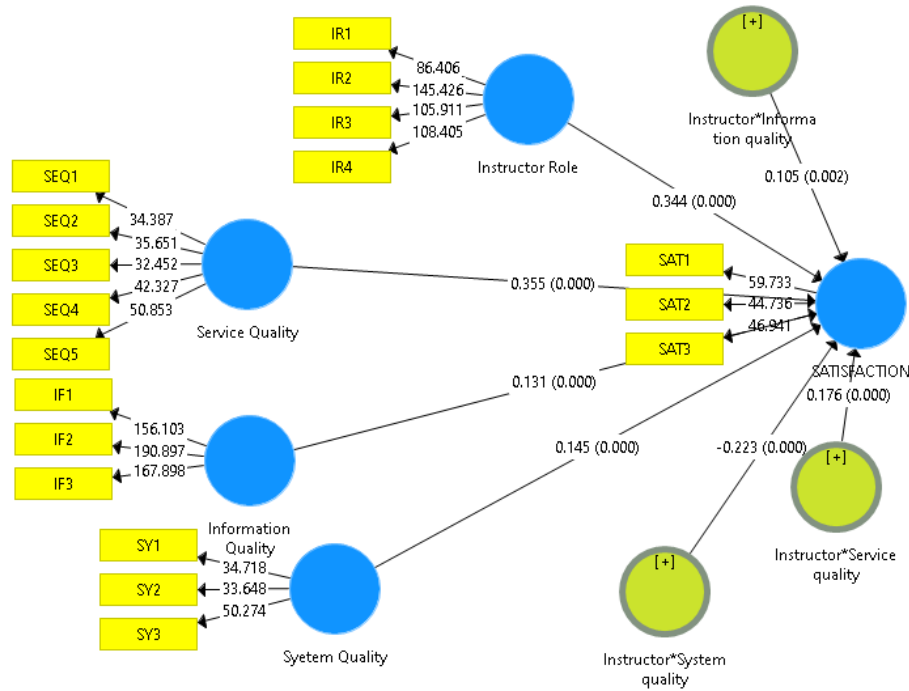


Figure 2. Results of path analysis



Figure 3. Slope plot

Table 7. Results of partial least squares structural (PLS) predict

Indicators	PLS Q ² Predict	PLS MAE	LM MAE
SAT1	0.534	0.504	0.471
SAT2	0.378	0.59	0.619
SAT3	0.439	0.559	0.573

Note: Q²_{predict} = predictive relevance statistic; MAE = mean absolute error; LM = linear model

Prediction errors for the manifest variables (MVs) of the endogenous construct learner satisfaction were analyzed. Figure 4 presents PLS errors for all items of learner satisfaction which shows non-symmetrical distribution of the errors. If the distribution of the errors is not normal, a comparison is made between PLS Mean Absolute Error (MAE) values and Linear Model (LM) MAE values for all the items of dependent construct. The same is done and statistics are presented in Table 7. MAE values for PLS model are smaller than the MAE values for LM model in two out of three variables. This suggests that the PLS model typically exhibits lower prediction errors than the linear benchmark. According to Shmueli et al. (2019) guidelines, these results show moderate predictive relevance, implying that even though the model does not show strong predictive superiority across all indicators, it still has a meaningful and practically useful predictive capability in terms of explaining and forecasting learner satisfaction.

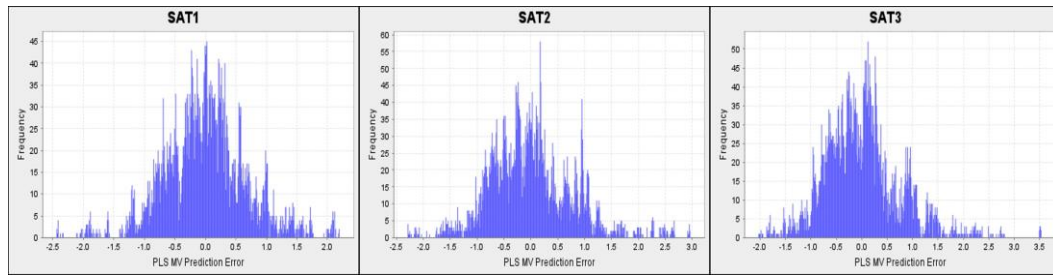


Figure 4. Prediction errors for the manifest variables of the endogenous construct learner satisfaction

4. Discussion

EdTech has acted as a catalyst for multiple transformations in classroom teaching and learning through digital solutions (Sharma, 2022). The study uses DeLone and McLean IS Success Model, 2003 to examine the impact of service quality of EdTech learning system on learner satisfaction. The model was extended by incorporating role of teacher as moderating construct in facilitating the relationship between service quality and learner satisfaction. This will help in exploring the learner's psychological position of being implicated in the e-learning system because of teacher's role in managing e-learning system, course materials quality, e-learning administrative as well as support service quality (Pham et al., 2019).

Based on the findings, H1 was supported as the results revealed information quality to be a significant predictor of learner satisfaction, which is in consonance with the findings of the prior studies (Eppler et al., 2003; Jun & Kang, 2013; Kim & Lim, 2001). This indicates that if the information quality provided in EdTech platform is precise, legitimate, up-to-date, free of mistake, and correctly presented, learners will be in a better position to understand the information, and thus, his satisfaction level will increase towards the EdTech platform.

Furthermore, regarding the relationship between service quality and learner satisfaction, the results are consistent with previous studies (Htang, 2021; Ibrahim et al., 2012; Jing & Yoo, 2013; Lee, 2010; Salameh et al., 2018; Tajuddin et al., 2013) and thus, H2 stands accepted. It is corroborated that e-learning service quality (including dimensions like online assistance, functionality, interactivity, response to service requests and quick solution to complaints) plays a significant role in helping EdTech platforms retain learners in a competitive education market (Yeo, 2009).

H3 was supported, as the results revealed that system quality positively influenced learner satisfaction. This suggests that desirable system characteristics, such as user-friendliness, interaction between the user and the platform, and speed of information access, are important attributes shaping satisfaction with an e-learning system. (Dubey et al., 2012; Dreheeb et al., 2016). A reliable, easy-to-use, and accessible e-learning system is therefore likely to enhance learner satisfaction.

In consonance with prior studies (Htang, 2021; Ibrahim et al., 2012), results from this study also supported the association between teacher's role and learner satisfaction and hence H4 was accepted. This reveals that teacher-related factors, such as enthusiasm, responsiveness to queries, individualized consideration, and the ability to motivate learners to exert additional effort, also affects learner satisfaction level as it aids in better understanding on EdTech Platforms.

The main objective of the paper was to analyze the moderating role of teacher in affecting learner satisfaction with system, service and information quality of EdTech Platforms. Results of the study accepted H5 and validated teacher's role in moderating the relationship between information quality and learner satisfaction as findings indicated positive interaction effects of teacher's role on information quality and learner satisfaction.

The study also accepted H6 as it found positive interaction impact of teacher's role on the relationship between service quality and learner satisfaction. There is a dearth of literature attempting to examine this hypothesized relationship. These findings may be attributed to the fact that EdTech platforms place teachers at the center of e-learning services, including course material management, interaction and coordination with learners, feedback, and evaluation, thereby increasing learners' attention to the role of teachers in delivering e-learning services.

Furthermore, the results also pinpointed negative but significant interaction effect of teacher's role on the relationship between system quality and satisfaction with EdTech Platforms. This signifies that when teacher involvement is high, the positive impact of system quality on learner satisfaction tends to weaken. The substitution effect between technological functionality and human support could be one reason for this finding. Students may rely less on the platform's technical features, usability, or performance efficiency when teachers actively clarify, explain, and customize learning experiences. Therefore, when there is robust human support available, the relative importance of system quality decreases.

Another possible mechanism relates to the rigidity of platform systems. The limited flexibility and standardized

structures of many EdTech systems may make it difficult to tailor them to the needs of specific learners. On the other hand, teacher intervention introduces adaptability, contextual explanation, and emotional support, thereby compensating for technological limitations. In these hybrid environments, pedagogical engagement may have a greater influence on user satisfaction than system functionality alone. This contradicted the findings of the study conducted by Topal (2016) which posited that satisfaction level of the students increases with the increase in use of various learning management system by the teachers like survey, virtual classroom, animation, video, forum, chat, and email. The marginal contribution of system quality to satisfaction may decrease if teachers encourage alternatives to the system's functionality rather than complementing it. In order to maximize learner satisfaction in EdTech environments, future research in learning management system (LMS) and blended learning contexts should further investigate the balance between technological robustness and adaptive teacher intervention.

5. Theoretical Implications

The study adds to the body of knowledge on IS success by introducing the teacher as a crucial boundary condition in determining students' satisfaction and by expanding the applicability of the DeLone and McLean IS Success Model to the context of K-12 digital learning environments. The study advances existing theory by validating the fundamental constructs of information quality, system quality, and service quality in a population that is still underrepresented in IS research i.e. school children in grades 5 through 12. Further, the study makes a valuable contribution in the existing literature by adding teacher as moderator to identify whether the impact of service quality, information quality and system quality on learner satisfaction becomes stronger in the presence of teacher as moderator.

The study conceptualizes digital learning as a socio-technical system in which human facilitation can either amplify or dampen learners' perceived satisfaction. Human support can offset technological complexity and shift learner's evaluation focus from interface features to pedagogical usefulness. This study goes beyond previous validation studies that primarily examined the functionality of the model's measures and relationships. Rather, it transforms the model by demonstrating that platform attributes and the role of the teacher work together to create learner satisfaction rather than just technology. This repositioning opens up new theoretical discussions about how interactions between pedagogical actors and technological features lead to user satisfaction.

6. Managerial Implications

The managerial implications of this study are fourfold. First, the results of this study help EdTech service providers to understand the satisfaction reported by students in e-learning using EdTech Platforms. They would be in a better position to pay greater attention in building their e-learning platforms in the illumination of various dimensions of learners' quality insight.

Secondly, the extended DeLone and McLean IS Success Model offers a greater insight into interrelationships of its various constructs and the moderating role of teacher in enhancing learner satisfaction with EdTech Platforms. As findings of the study indicated that teacher's role enhances the effects of information quality and service quality on learner satisfaction; therefore, teachers remain indispensable in the education system. It also implies that an effective e-learning environment is created by not only by system use but also by its capacity to improve learner satisfaction. Hence, EdTech service providers are urged to promote teacher's proficiency regarding information, service, and system quality. EdTech Platforms should organize workshops and training programs to support teachers in designing course structure and content to enhance user's learning outcomes.

Thirdly, the results persuade policy makers to stimulate the use of EdTech for e-learning as technological development coupled with greater accessibility has offered multiplex opportunities and ease to the learners. Though overall results of the study demonstrated that technology alone cannot be used as a surrogate to substitute human-to-human communiqué with computer-mediated interaction and learning tools, but significant finding was obtained from the study that teacher's presence weakens the impact of system quality on the satisfaction. This may be due to the reason that systems are designed in an inflexible manner leaving little space for customizations and modifications. On the contrary, human interventions can lead to customization of processes considering the needs of the learners. Hence, EdTech Platforms providing education to the students should be flexible and customizable, so that teacher can change it according to the need and level of the learner.

Finally, as the results of this study have underlined that learner satisfaction has been shaped by information quality, service quality, and system quality (DeLone and McLean IS Success Model) in e-learning environment – hence there should correspondingly be adequate support and training should therefore be provided for users to promote the adoption of e-learning on EdTech platforms.

7. Conclusions

As digital modes have become increasingly central to education, expectations regarding educational equity have

also become more closely tied to educational technology providers. To this end, this study examined the impact of service quality of EdTech Platforms on learner satisfaction by using DeLone and McLean IS Success Model, 2003. The study extended the model and added teacher's role as moderating variable. It then explored the role of teacher on EdTech platform in affecting the relationship between model constructs and satisfaction. Overall results of the study confirmed significant positive relationship between IS Success Model constructs and learner satisfaction. Also, moderating role of teachers has been established between DeLone and McLean IS Success Model constructs and learner satisfaction. The impact of service and information quality on learner satisfaction is enhanced in the presence of teacher whereas impact of system quality is decreased in teacher's presence. Hence, it can be concluded that teachers cannot be eliminated from EdTech Platforms. Human intervention is certainly required for making inflexible, pre-formatted EdTech systems work efficiently to enhance learner satisfaction.

8. Limitations of the Study

Although the study has resulted into both theoretical and practical additions to existing literature, but some limitations were unavoidable. Firstly, because the data were collected using the snowball sampling method, the results may have been affected in several ways. The sample may be homogeneous, reflecting similar traits, educational backgrounds, or platform preferences as participants were selected through referrals. This may restrict the diversity of responses and result in selection bias. Additionally, certain groups of students (e.g., those less socially connected or less active on EdTech Platforms) may have been underrepresented, affecting the generalizability of the findings. Secondly, while carrying out research, only learner's perspective was covered, whereas teacher's perspective is equally important. Hence, future studies can focus on analyzing teacher's experience as well as satisfaction with the EdTech Platforms. Thirdly, the study conducted was cross-sectional, and learner's perceptions may change while using EdTech Platforms over time and this cannot be traced in conjunction with a cross-sectional study. These limitations provide ample scope for future research. As technology is evolving swiftly, it will be advantageous to substantiate the results in longitudinal settings to investigate in what way technological innovation affects learner satisfaction with EdTech Platforms.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflict of interest.

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